INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

16 April 2025

Subject CS1 – Actuarial Statistics
Core Principles

Paper B

Time allowed: One hour and fifty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

All material in this document is the copyright material of the Institute and Faculty of Actuaries. No part of this document may be reproduced in any material form, whether by publication, translation, storage in a retrieval system or transmission by electronic, mechanical, photocopying, recording or other means, without the prior permission of the Institute and Faculty of Actuaries.

1 A consultancy firm has created an index, denoted by I, used to summarise the health status of individuals in a single number. They have asserted that this index can be used to predict annual healthcare costs, Y, using linear regression. As evidence for their assertion, they provide a sample of observed healthcare costs y_i, and the predicted values $\hat{y}_i = \hat{a}I_i + \hat{b}$. For data protection and commercial reasons, they do not provide the values of I_i or the estimated values $\hat{a}$ and $\hat{b}$.

This dataset is saved in the file 'CS1B HealthCareIndex.Rdata'. After loading the data into R, using the command `load("CS1B HealthCareIndex.Rdata")`, the following two variables will be available:

- `Predicted_costs` – the predicted annual healthcare costs, $\hat{y}_i$, calculated as $\hat{y}_i = \hat{a}I_i + \hat{b}$ using the company's healthcare index.
- `Realised_costs` – the observed annual healthcare costs y_i.

(i) Plot the residuals of the linear regression model against the predicted annual healthcare costs, $\hat{y}_i$. [4]

(ii) Suggest, with reasons, what can be concluded from the plot in part (i). [2]

(iii) Determine the correlation coefficient between the predicted annual healthcare costs, $\hat{y}_i$, and the realised annual healthcare costs, y_i, based on the provided data. [2]

(iv) Determine an estimate of the correlation coefficient between the healthcare index, I, and the realised annual healthcare costs, Y, based on the provided data. [3]

[Total 11]

2 A racing car driver is interested in the relationship between the speed of a car and its stopping distance (the distance required for the car to come to a complete stop after applying the brakes). They have carried out a series of tests of the stopping distance at different speeds. This data is recorded in the file 'CS1B StoppingDistance.RData'.

After loading the data into R, using `load("CS1B StoppingDistance.RData")`, the following two variables will be available:

- `speed` – the speed of the car (in kilometres per hour) when the driver applied the brakes.
- `stopping_dist` – the stopping distance (in metres).

(i) Fit a linear model to the data, with `stopping_dist` as the response variable and `speed` as the explanatory variable. [2]

(ii) Plot, on the same graph:

(a) The sample quantiles of the residuals of the model fitted in part (i) against the theoretical Normal quantiles.

(b) A line that passes through the first and third quartiles of the quantile-quantile plot in part (ii)(a).

[**Hint:** Use the `qqline()` function.]

[5]

(iii) Plot the residuals of the model fitted in part (i) against the explanatory variable. [3]

(iv) Comment on the suitability of the model fitted in part (i), based on the plots produced in parts (ii) and (iii). Your answer should include suggested improvements where appropriate. [6]

[Total 16]

3 A random variable is assumed to follow a uniform distribution: $X \sim U(0.5, 2.5)$.

(i) Calculate the mean of X. [1]

Use the command `set.seed(7)` to initialise the random number generator.

(ii) (a) Perform two separate simulations of X, one for 10 and then one for 10,000 simulations.

You should store these simulations as `sim_x1` and `sim_x2` respectively, for later use. [4]

(b) Estimate, for each of the two simulations in part (ii)(a), the mean of X. [3]

(c) Comment on your results from part (ii)(b) and part (i). [2]

(iii) (a) Plot, on the same graph, the density of X based on the 10,000 simulated values from part (ii)(a) and the theoretical probability density function of X.

[**Hint:** You may find the following R commands useful: `density()` and `curve()`.]

[8]

(b) Comment on your plot in part (iii)(a). [2]

[Total 20]

4 Let $X_1, \ldots, X_n$ be independent identically distributed random variables with a density function given by:

$$f(x,\theta) = \begin{cases} \theta x^{\theta-1}, & x \in [0,1] \\ 0, & \text{otherwise} \end{cases}$$

where $\theta > 0$.

You are given that if X follows the distribution defined above, $Y = \log(X)$ follows an exponential distribution with parameter θ.

(i) Generate $n = 20$ random values from the exponential distribution with parameter 5. Use the command `set.seed(222)` to initialise the random number generator before you start the simulation. [1]

(ii) Generate $n = 20$ random values from the distribution of X with probability density function $f(x, \theta)$ as given above, when $\theta = 5$, using your answer in part (i). [2]

You are given four different estimators of θ: the method of moments estimator $\hat{\theta}_A$, the maximum likelihood estimator $\hat{\theta}_B$, a Bayes estimator $\hat{\theta}_C$ and a further estimator $\hat{\theta}_D$, such that:

$$\hat{\theta}_A = \frac{\bar{X}}{1-\bar{X}}$$

$$\hat{\theta}_B = -\frac{n}{\sum_{i=1}^{n} \log(X_i)}$$

$$\hat{\theta}_C = \frac{2(n+1)}{1-2\sum_{i=1}^{n} \log(X_i)}$$

$$\hat{\theta}_D = -\frac{n-1}{\sum_{i=1}^{n} \log(X_i)}$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

(iii) (a) Perform a simulation to obtain 1,000 values of each of the four estimators when $n = 20$ and $\theta = 5$. Use the command `set.seed(222)` to initialise the random number generator before you start the simulation. [7]

(b) Calculate the mean square error estimate for each of these estimators based on part (iii)(a). [13]

(c) Comment on which of the four estimators appears to be the most efficient using part (iii)(b). [2]

[Total 25]

5 An estate agent wants to understand what factors affect the sale price of properties in their area. They have collected data for 20 houses sold recently.

Use the command `load("CS1B_houses.RData")` to load the observed data into R. This will create the following variables in R:

- `days` to agree sale
- `price` of the property, in thousands
- `sqft` the area (in square feet) of the property
- `bedrooms` indicating if the property has more than three bedrooms (1 = yes, 0 = no).

(i) Fit a Generalised Linear Model (GLM), using a normal distribution, to the sale `price` with `days`, `sqft` and `bedrooms` as the main effect explanatory variables. Use the identity function as the link function. Your answer should include a summary of the fitted model. You should save this model as `Model1`. [4]

(ii) Comment on the significance of the parameters of the model fitted in part (i). [4]

(iii) Fit a second GLM as in part (i), but excluding `days` from the covariates, showing the summary. You should save this model as `Model2`. [3]

The estate agent is concerned that this model isn't suitable and would like to investigate the impact of the covariates further.

(iv) Plot the observed sale price against the square footage `sqft` for properties with at least three bedrooms and for properties with fewer than three bedrooms using two different colours to distinguish the properties with respect to the number of bedrooms. [3]

(v) Comment on your plot in part (iv) focussing on the relationship between square footage and sale price. [2]

The estate agent also has doubts about the suitability of the normal distribution and the identity link function.

(vi) Fit a third model using the gamma distribution with its canonical link function, and `sqft` and `bedrooms` as the main effect explanatory variables. You should save this model as `Model3`. [3]

(vii) Comment on whether `Model2` from part (iii) or `Model3` from part (vi) is a better fit, based on the Akaike information criterion. [2]

(viii) Plot the residuals of `Model2` from part (iii) against the fitted values. [3]

(ix) Comment on the residuals plotted in part (viii). [2]

(x) Calculate the predicted sale price using `Model2` from part (iii) of a new property about to be listed. The new property has two bedrooms and its area is 1,500 square feet. [2]

[Total 28]

END OF PAPER